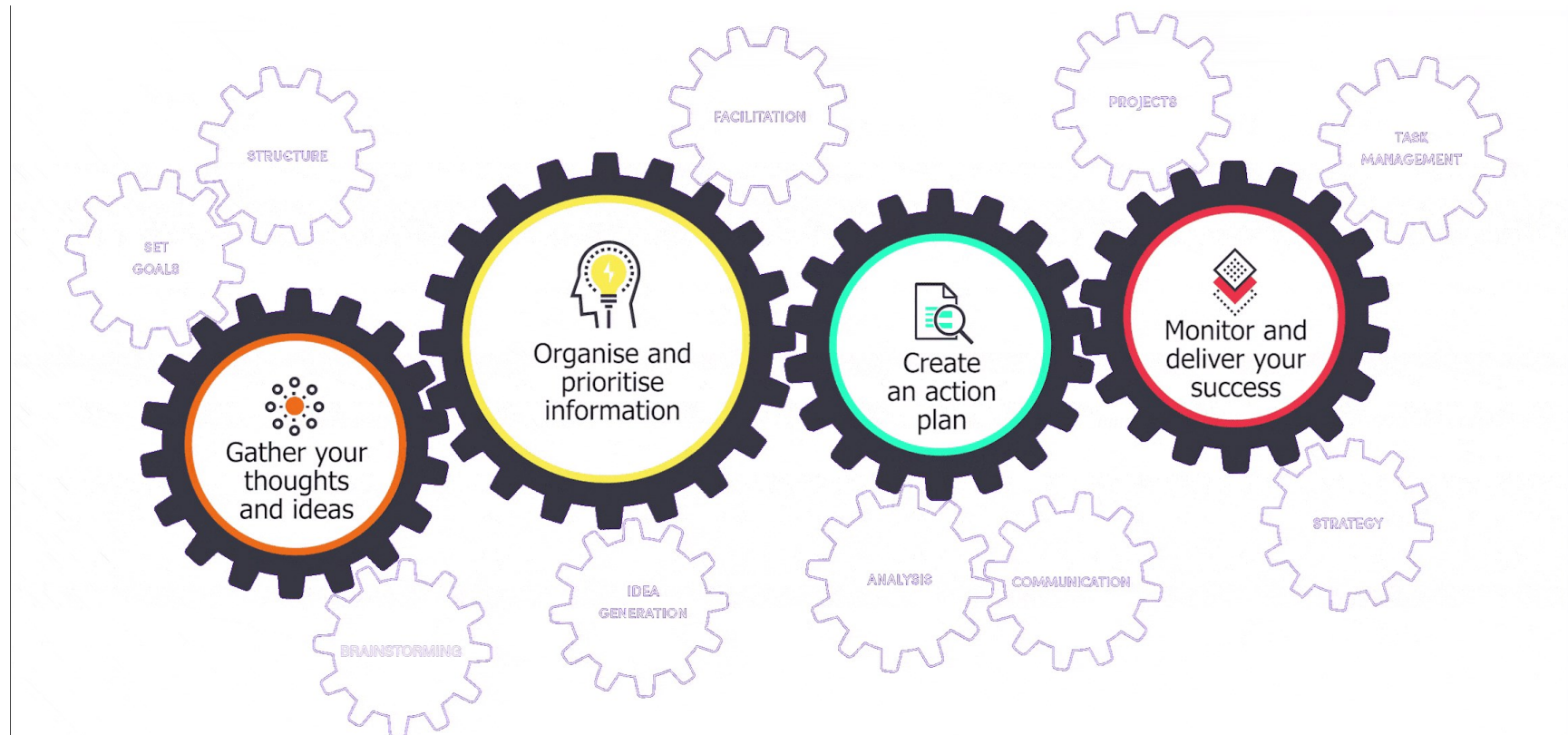


# UNIT-5

## SOFTWARE ENGINEERING & PROJECT MANAGEMENT

# Software Management Goals

- The main goal of software project management is to enable a group of developers to work effectively towards the successful completion of a project.



## SOFTWARE PROJECT MANAGEMENT COMPLEXITIES

- ▶ **Invisibility:** Invisibility of software makes it difficult to assess the progress of a project and is a major cause for the complexity of managing a software project.
- ▶ **Changeability:** Frequent changes to the requirements and the invisibility of software are possibly the two major factors making software project management a complex task.
- ▶ **Complexity:** Even a moderate sized software has millions of parts (functions) that interact with each other in many ways—data coupling, serial and concurrent runs, state transitions, control dependency, file sharing, etc.
- ▶ **Uniqueness:** Every software project is usually associated with many unique features or situations. This makes every project much different from the others. This is unlike projects in other domains, such as car manufacturing or steel manufacturing where the projects are more predictable. Due to the uniqueness of the software projects, a project manager in the course of a project faces many issues that are quite unlike the ones he/she might have encountered in the past.
- ▶ **Exactness of the solution:** Mechanical components such as nuts and bolts typically work satisfactorily as long as they are within a tolerance of 1 per cent or so of their specified sizes. However, the parameters of a function call in a program are required to be in complete conformity with the function definition.
- ▶ **Team-oriented and intellect-intensive work:** Software development projects are akin to research projects in the sense that they both involve team-oriented, intellect-intensive work. In contrast, projects in many domains are labor-intensive and each member works in a high degree of autonomy. Examples of such projects are planting rice, laying roads, assembly line manufacturing, constructing a multistoried building, etc.

# Types of Management in Software Project

## 1. Conflict Management

Conflict management is the process to restrict the negative features of conflict while increasing the positive features of conflict. The goal of conflict management is to improve learning and group results including efficacy or performance in an organizational setting. Properly managed conflict can enhance group results.

## 2. Risk Management

Risk management is the analysis and identification of risks that is followed by synchronized and economical implementation of resources to minimize, operate and control the possibility or effect of unfortunate events or to maximize the realization of opportunities.

## 3. Requirement Management

It is the process of analyzing, prioritizing, tracking, and documenting requirements and then supervising change and communicating to pertinent stakeholders. It is a continuous process during a project.

# Types of Management in Software Project

## 4. Change Management

Change Management is a systematic approach to dealing with the transition or transformation of an organization's goals, processes, or technologies. The purpose of change management is to execute strategies for effecting change, controlling change, and helping people to adapt to change.

## 5. Software Configuration Management

Software configuration management is the process of controlling and tracking changes in the software, part of the larger cross-disciplinary field of configuration management. Software Configuration Management includes revision control and the inauguration of baselines.

## 6. Release Management

Release Management is the task of planning, controlling, and scheduling the built-in deploying releases. Release management ensures that the organization delivers new and enhanced services required by the customer while protecting the integrity of existing services.

# Aspects of Software Project Management

## 1. Planning

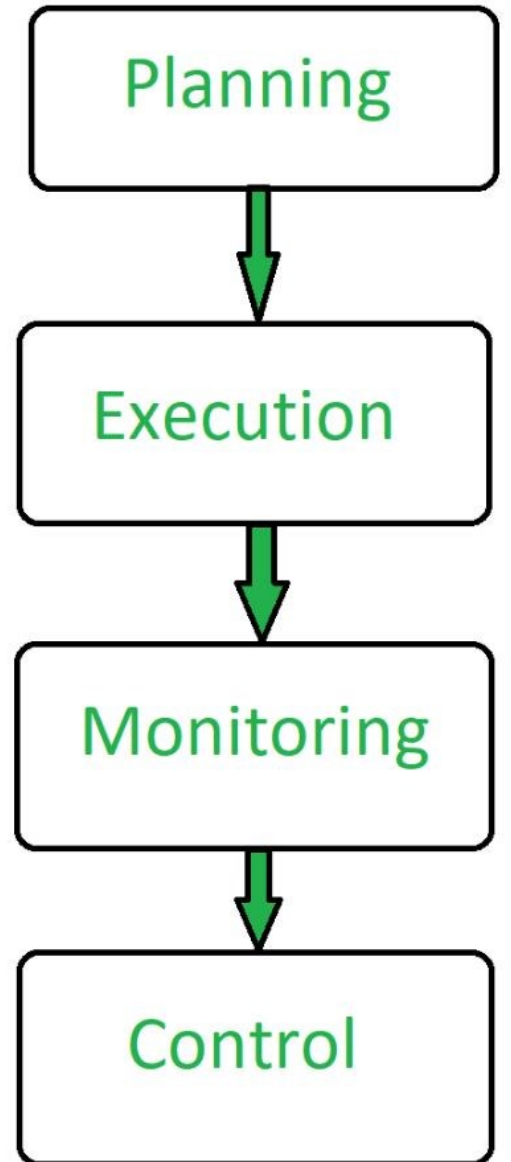
The Software Project Manager lays out the complete project's blueprint. The project plan will outline the scope, resources, timelines, techniques, strategy, communication, testing, and maintenance steps. SPM can aid greatly here.

## 2. Leading

A software project manager brings together and leads a team of engineers, strategists, programmers, designers, and data scientists. Leading a team necessitates exceptional communication, interpersonal, and leadership abilities. One can only hope to do this effectively if one sticks with the core SPM principles.

## 3. Execution

SPM comes to the rescue here also as the person in charge of software projects (if well versed with SPM/Agile methodologies) will ensure that each stage of the project is completed successfully. measuring progress, monitoring to check how teams function, and generating status reports are all part of this process.



# Aspects of Software Project Management

## 4. Time Management

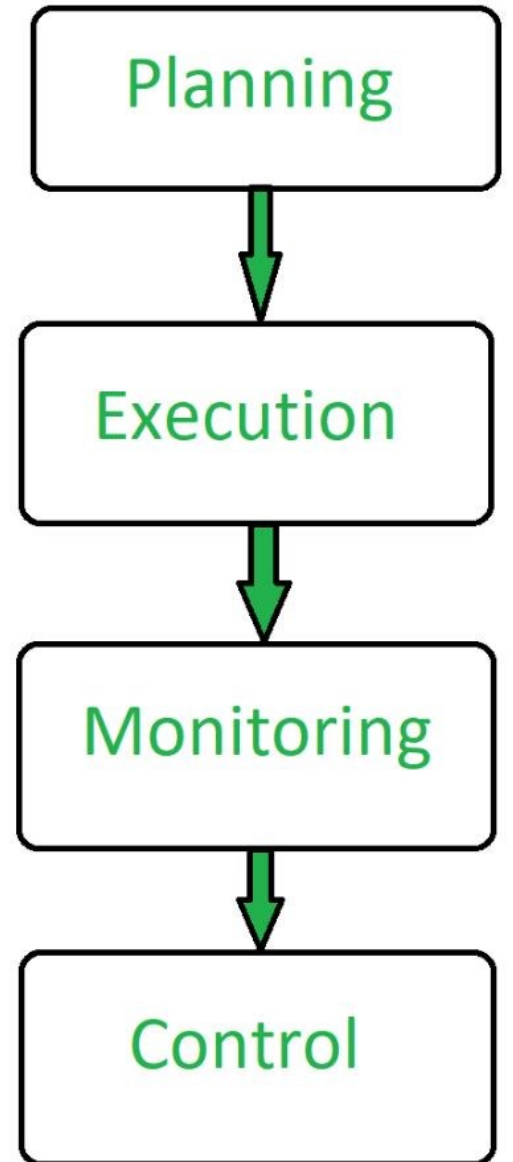
Abiding by a timeline is crucial to completing deliverables successfully. This is especially difficult when managing software projects because changes to the original project charter are unavoidable over time. To assure progress in the face of blockages or changes, software project managers ought to be specialists in managing risk and emergency preparedness. This Risk Mitigation and Management is one of the core tenets of the philosophy of SPM.

## 5. Budget

Software Project Managers, like conventional project managers, are responsible for generating a project budget and adhering to it as closely as feasible, regulating spending, and reassigning funds as needed. SPM teaches us how to effectively manage the monetary aspect of projects to avoid running into a financial crunch later on in the project.

## 6. Maintenance

Software project management emphasizes continuous product testing to find and repair defects early, tailor the end product to the needs of the client, and keep the project on track. The Software Project Manager makes ensuring that the product is thoroughly tested, analyzed, and adjusted as needed. Another point in favor of SPM.



# **The 10 Pillars of Perfect Projects**

One framework. Zero confusion.  
Let's build a banking app.

**By Arham Faraaz, PMP, PMI-ACP, CSM, SAFe RTE.**



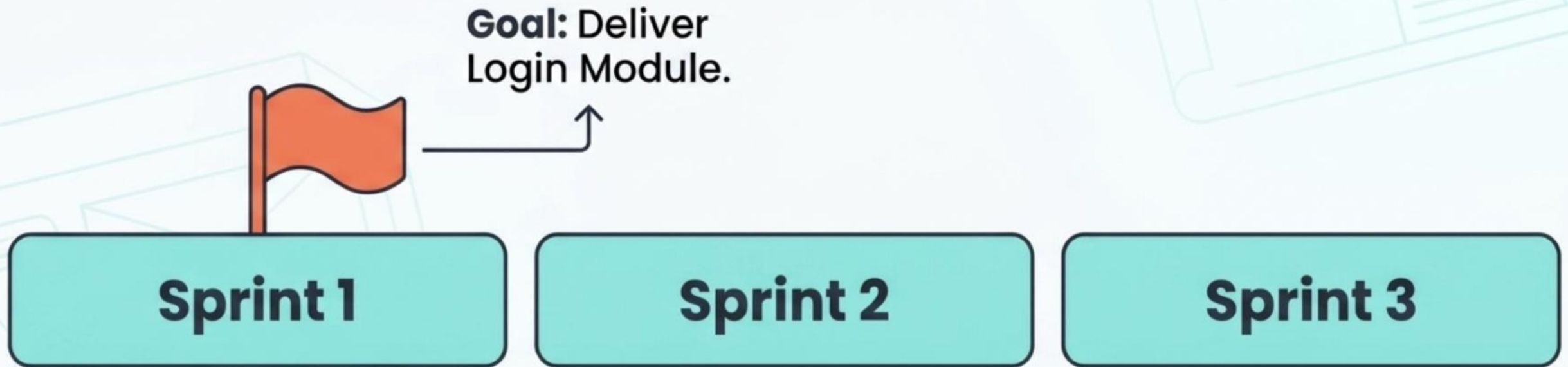
# 1. Scope Management Defines the Boundaries

Define exactly what is included and excluded. Without strict borders, your project will suffer from scope creep.



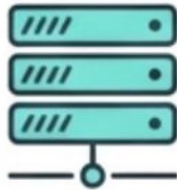
## 2. Schedule Management Controls the Clock

Plan your timelines, milestones, and deadlines to keep the team moving forward using tools like Gantt charts and sprint plans.



# 3. Cost Management Fuels the Engine

Estimate and control the burn rate. You are budgeting for tools, infrastructure, and human talent.



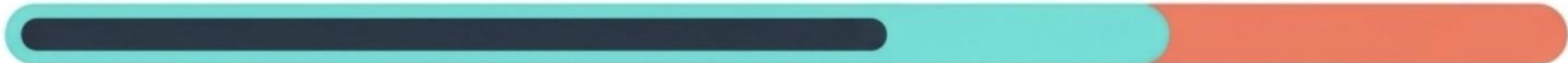
**Cloud Hosting**



**Developer Salaries**

**Budget Burn**

60% SPENT



**DANGER ZONE**

# 4. Quality Management Sets the Standard

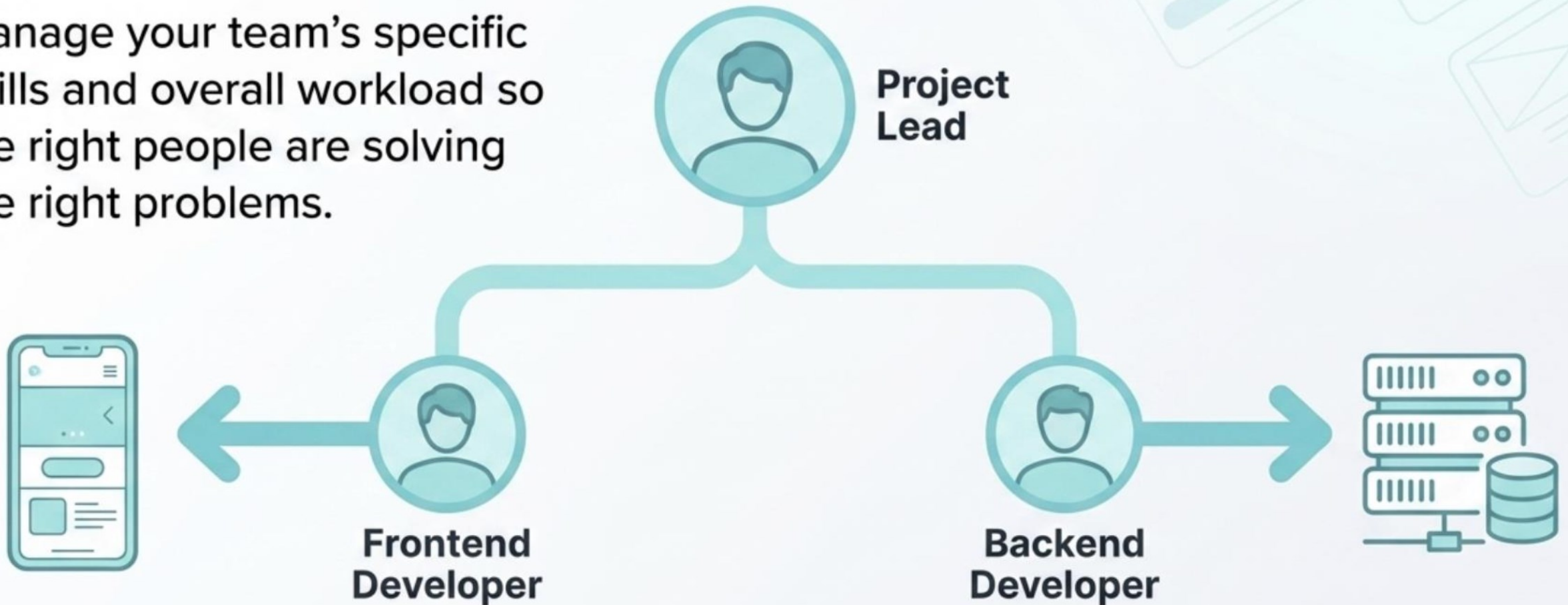
Ensure the product meets all requirements and standards through rigorous testing and QA processes.



**Target:** Zero critical bugs before release.

# 5. Resource Management Deploys the Talent

Manage your team's specific skills and overall workload so the right people are solving the right problems.



# 6. Communication Management Maintains the Pulse

Guarantee the exact right information reaches your stakeholders at the exact right time.

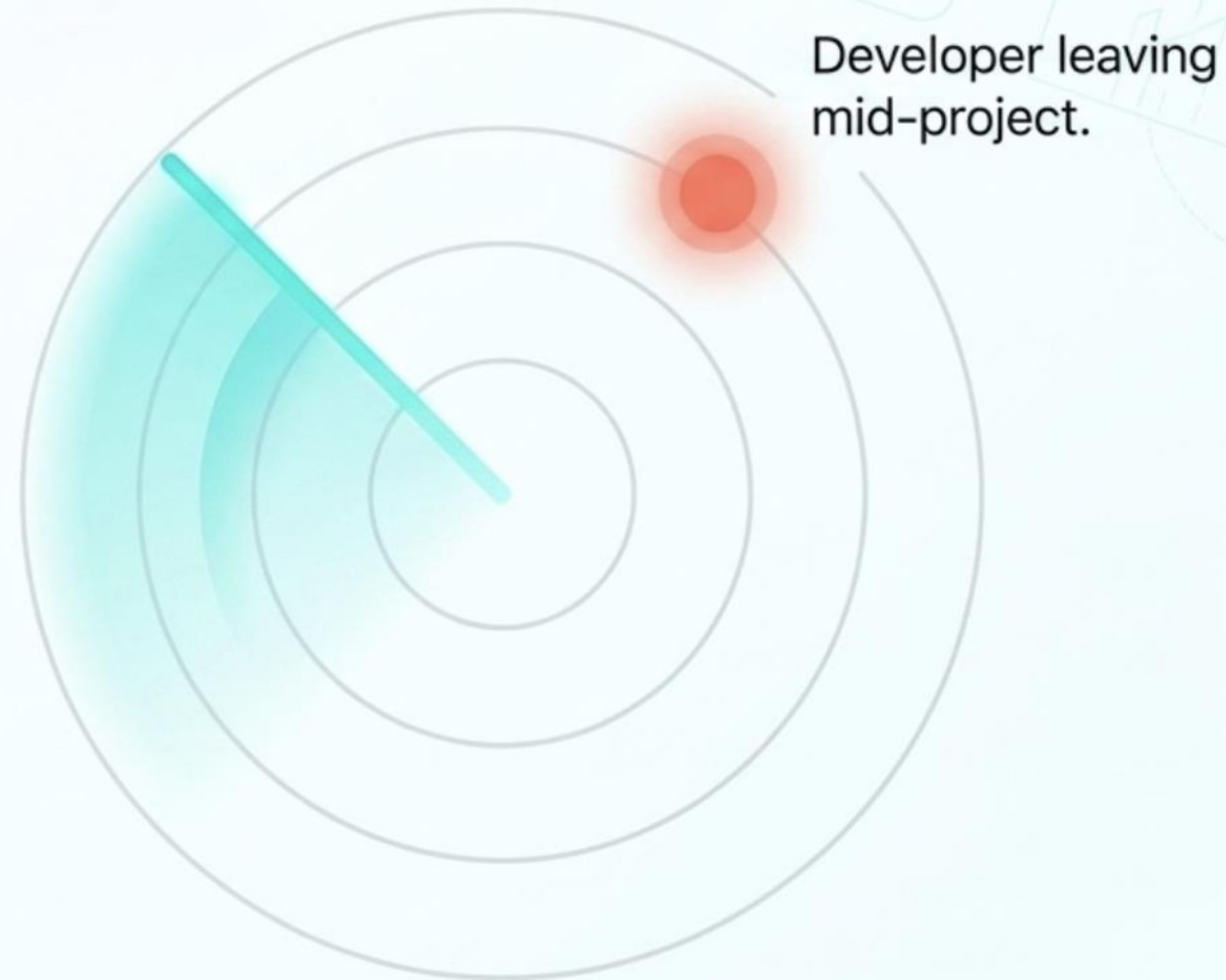


# 7. Risk Management Maps the Unknowns

Identify and mitigate threats long before they derail your timeline or budget.



**Mitigation:** Code documentation and cross-training.



## 8. Procurement Management Recruits Outside Help

Manage vendors, specialized tools, and third-party services when your internal resources aren't enough.

**Internal  
Core Team**



**External  
UI/UX Agency**

## 8. Procurement Management Recruits Outside Help

Manage vendors, specialized tools, and third-party services when your internal resources aren't enough.

**Internal  
Core Team**



**External  
UI/UX Agency**

# 9. Stakeholder Management Engages the VIPs

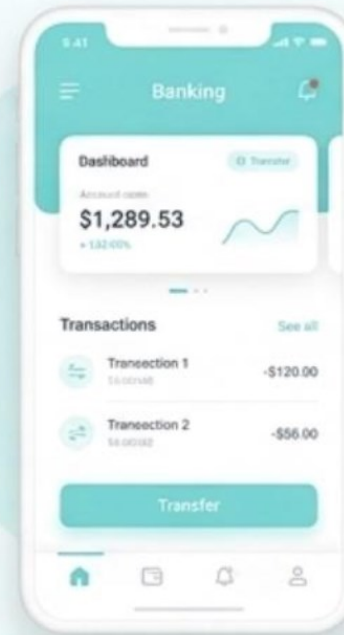
Identify and actively engage everyone who holds a vested interest in the success or failure of your project.



**The Client**



**Company Management**



**The End-User**

# 9. Stakeholder Management Engages the VIPs

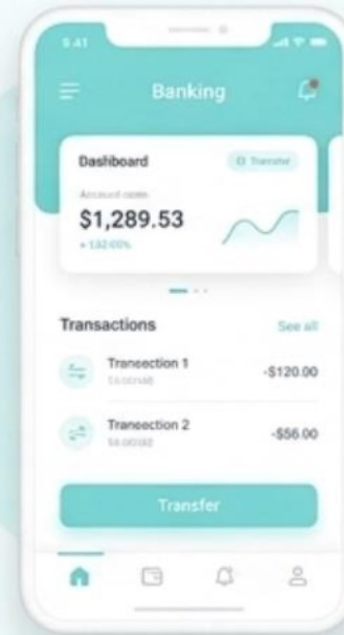
Identify and actively engage everyone who holds a vested interest in the success or failure of your project.



**The Client**



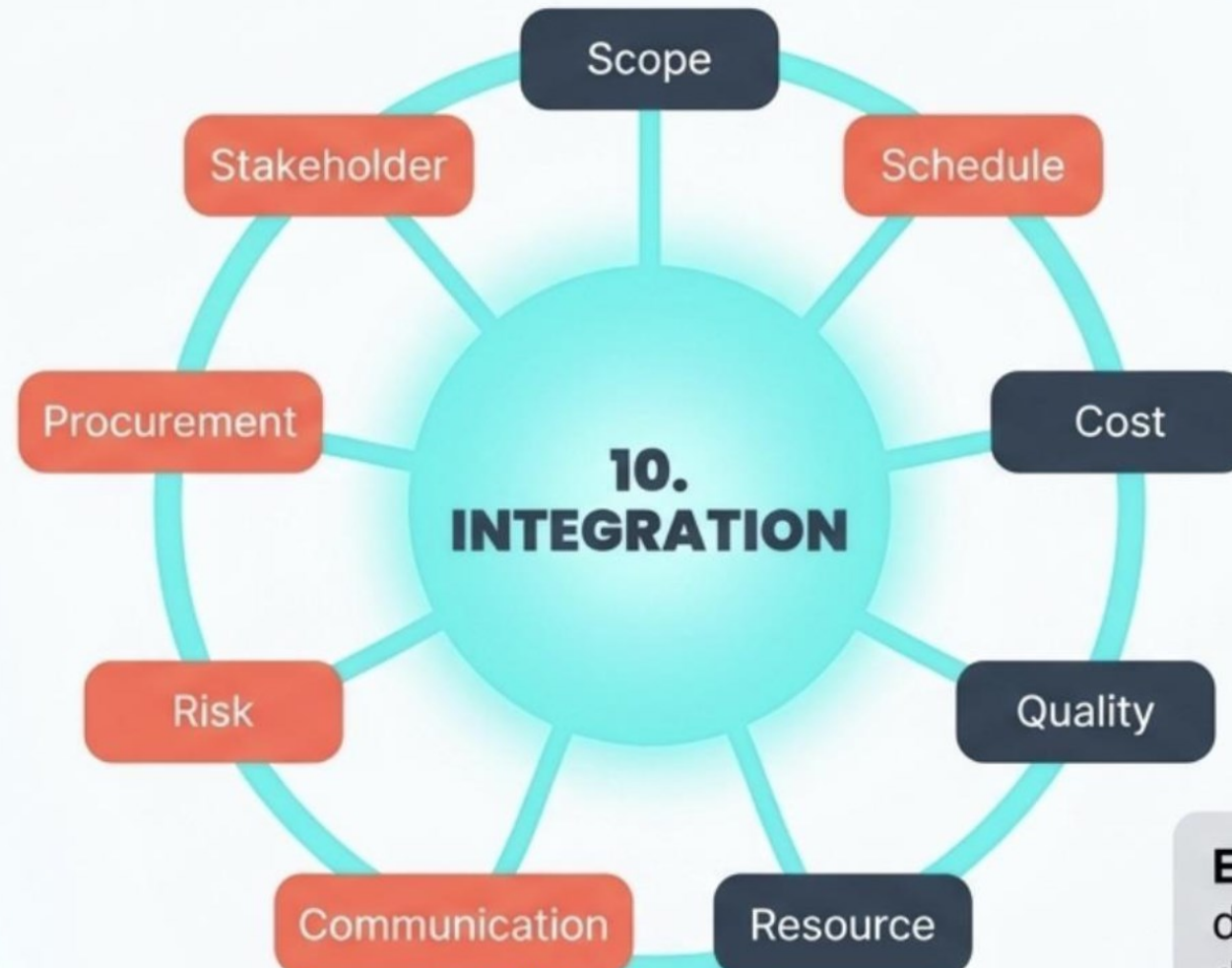
**Company Management**



**The End-User**

# 10. Integration Management is the Conductor

The ultimate unifier. Bringing everything together to ensure the project runs as a single, unstoppable system.

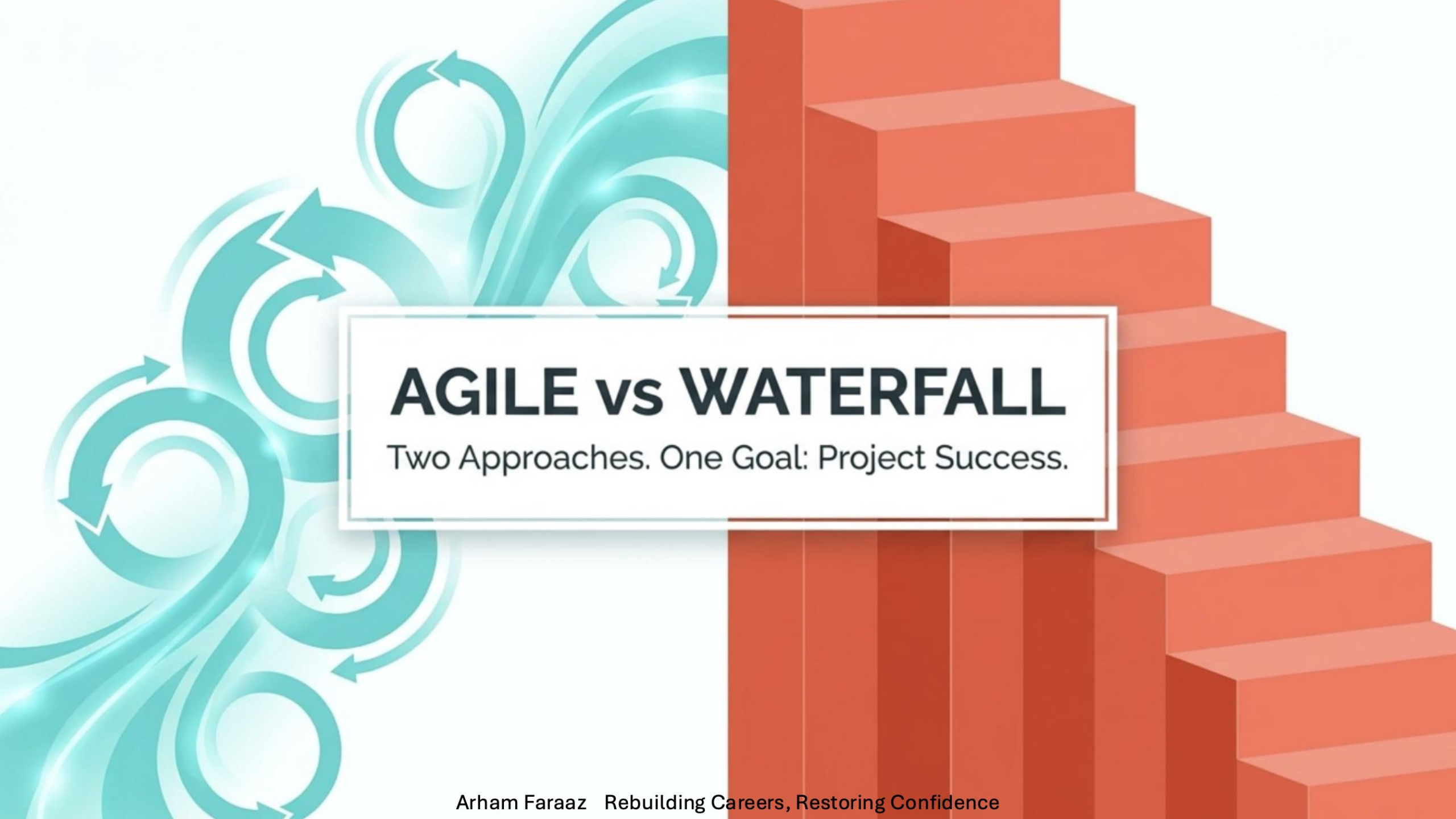


**Example:** Flawlessly aligning the development, QA testing, and live deployment of the banking app.

# The Master Blueprint Matrix

Screenshot this. Your ultimate cheat sheet for total project control.

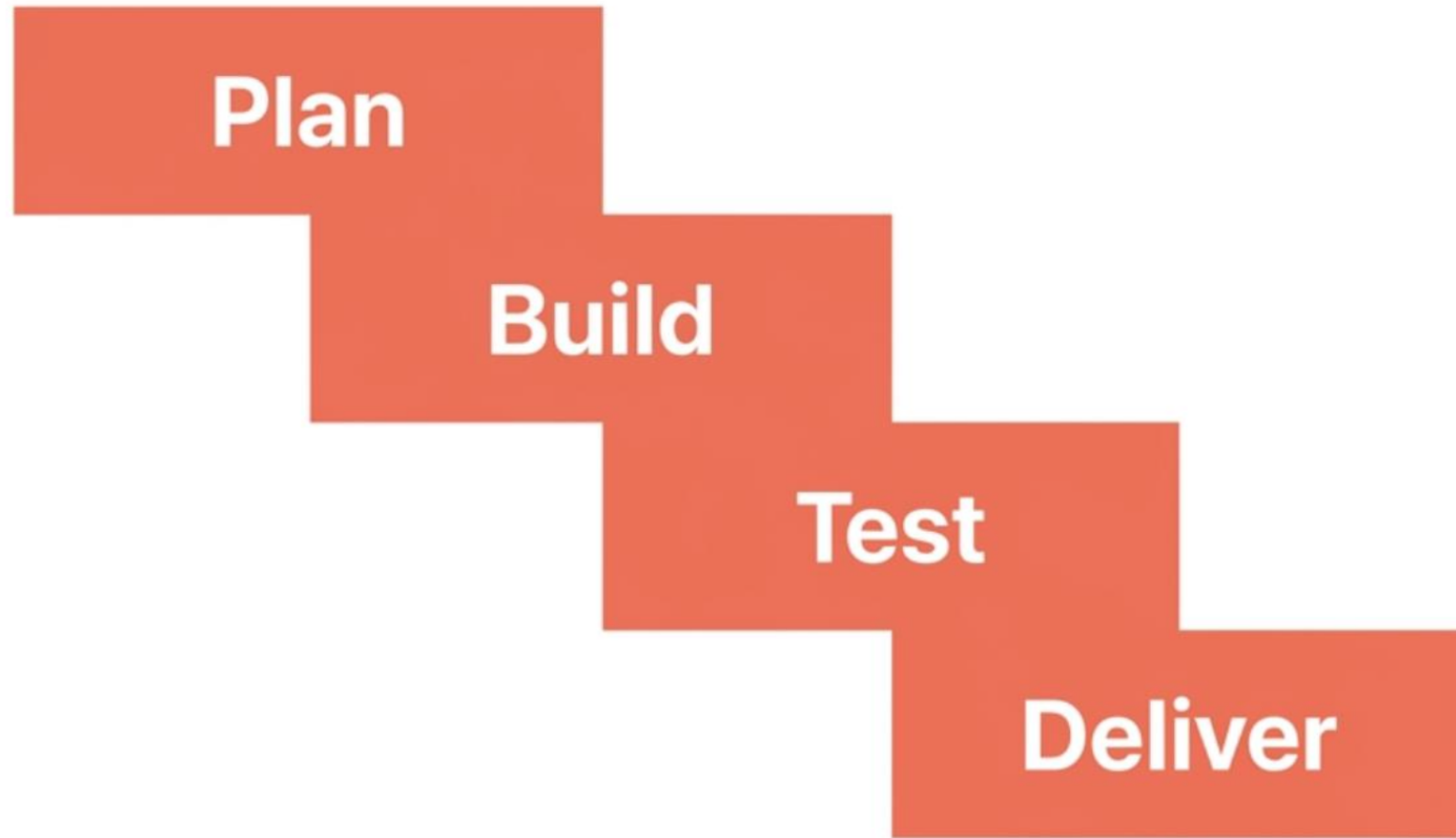
| The Area      | The Focus         | Banking App Example        |
|---------------|-------------------|----------------------------|
| Scope         | Boundaries        | Login/EMI features         |
| Schedule      | Timelines         | Sprint 1 deadlines         |
| Cost          | Budget            | Cloud hosting + Salaries   |
| Quality       | Standards         | Zero critical bugs         |
| Resource      | Talent            | Frontend/Backend devs      |
| Communication | Information       | Daily standups             |
| Risk          | Unknowns          | Dev leaving mid-project    |
| Procurement   | Outside Help      | Hiring external UI/UX      |
| Stakeholder   | The VIPs          | Client/Users/Management    |
| Integration   | The Single System | Aligning Dev/QA/Deployment |



# **AGILE vs WATERFALL**

Two Approaches. One Goal: Project Success.

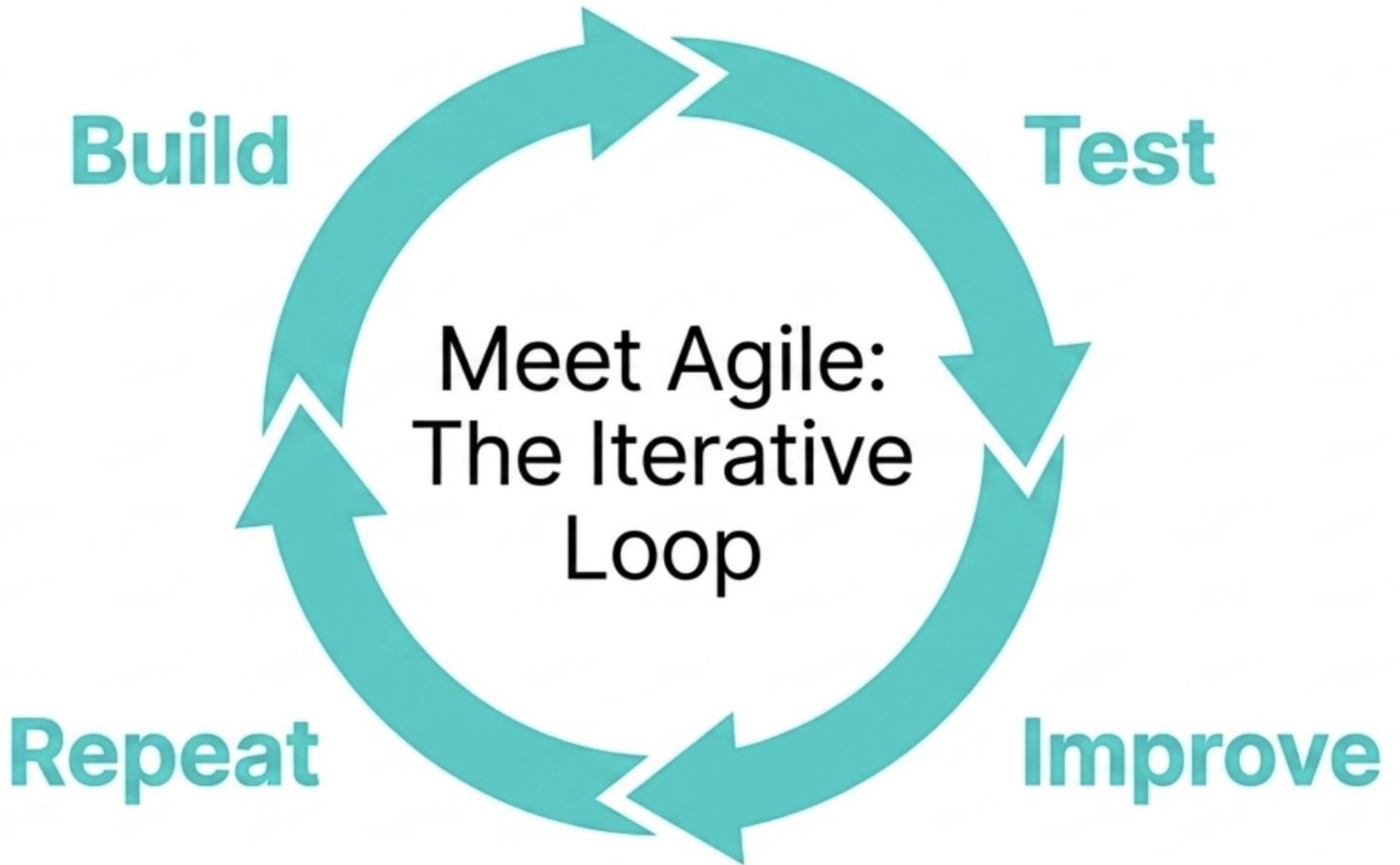
# Meet Waterfall: The Logical Sequence



**Step-by-step execution.**

**Everything planned upfront.**

**One final delivery.**





## Updating a Mobile App

Constant tweaks, user feedback, fast updates. Adapt to change.



## Building a Bridge

Measure twice, cut once, no going back.  
Follow the plan.

**Agile embraces change.**  
**Waterfall controls change.**

Choose wisely = Project Success.

Arham Faraaz Rebuilding Careers. Restoring Confidence

# The Agile Pulse : SPM

Principles of Value and Velocity.



# Navigating the Unknown

Complex

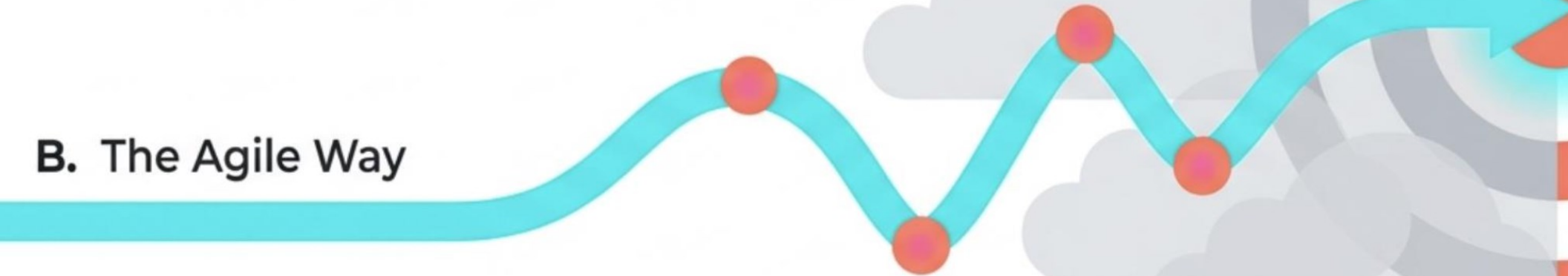
Uncertain

Varying Scope

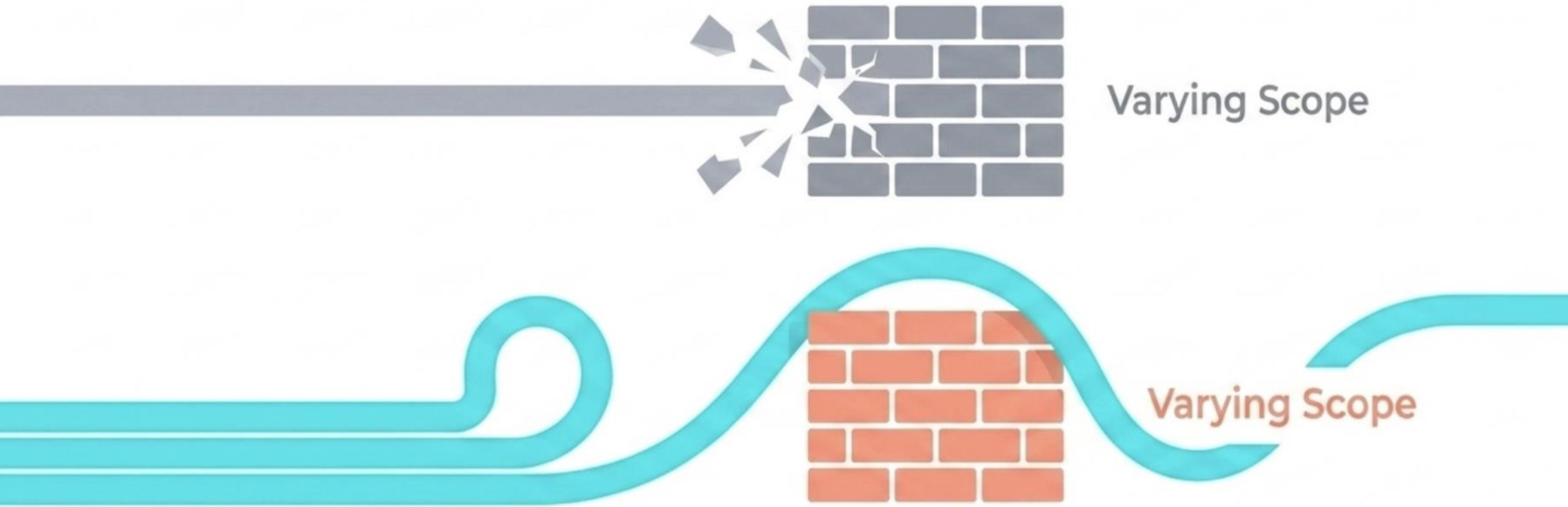
A. The Old Way



B. The Agile Way



# The Agile Way Of Working



Stop crashing into changing requirements. Start flowing with them.

# Upgrading How We Connect

## Face to Face Communication

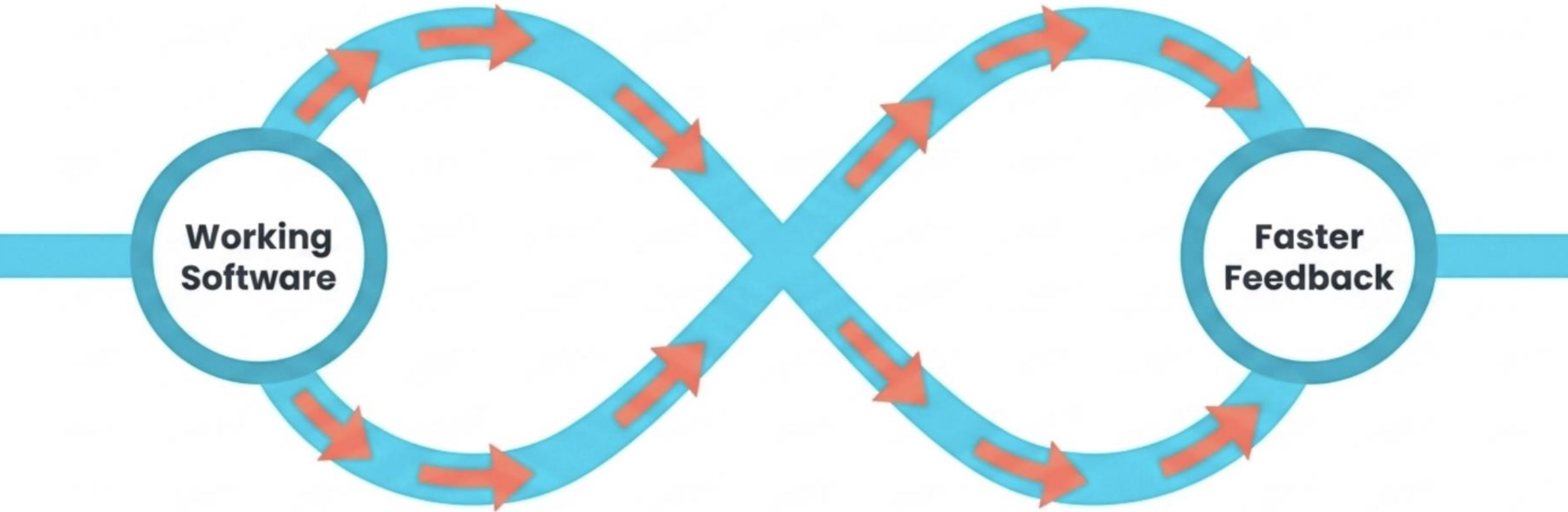
Slow & Misunderstood



Fast & Clear

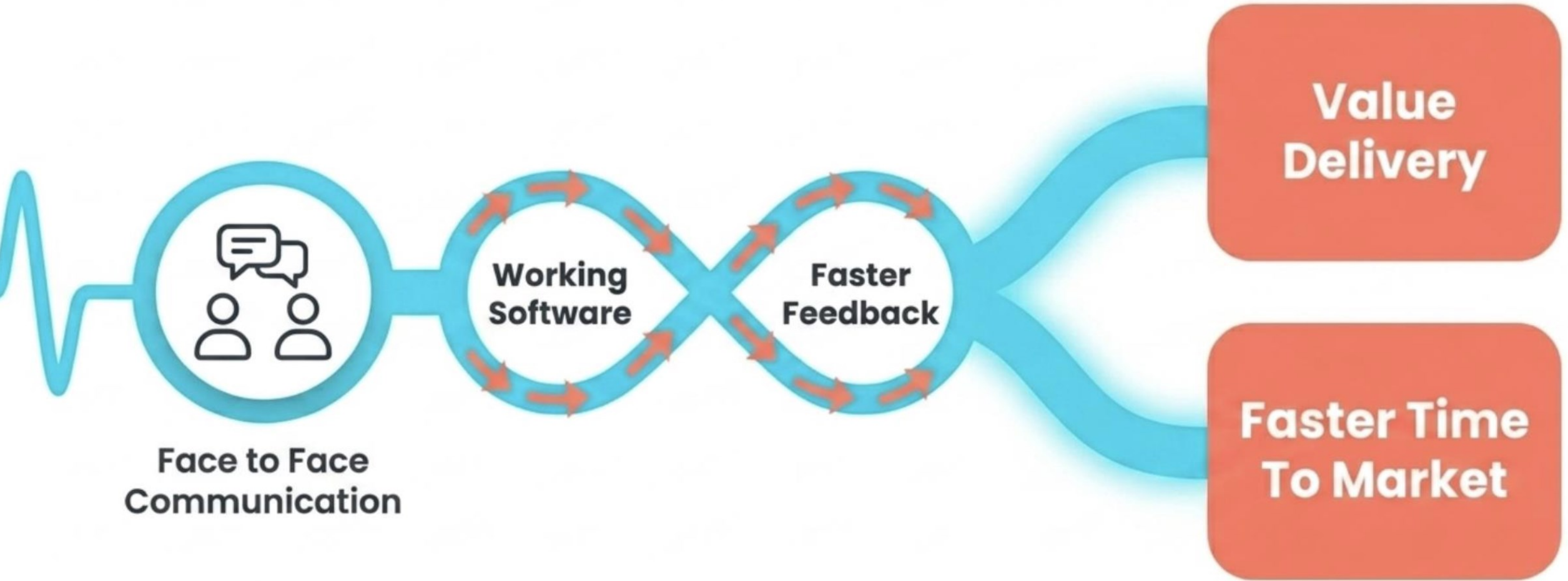


# The Engine of Iteration



Real progress isn't measured by plans. It's measured by what actually works, allowing you to course-correct instantly.

# The Ultimate Payoff



Speed isn't about rushing. Tighten the loop, communicate directly, and turn uncertainty into your greatest competitive advantage.

# 8 Reasons Software Projects Fail

## (And How to Spot Them)



# IT IS NEVER JUST “WRITING CODE”

## EXPECTATION

POINT A



POINT B

## REALITY

POINT A



POINT B

We all want a straight line, but modern development is a maze of moving targets and human friction.

# THE 3 DANGER ZONES OF PROJECT MANAGEMENT

**PROCESS**  
The Planning Traps



**PEOPLE**  
The Human Element



**EXECUTION**  
The Reality Check



A modern office interior with glass walls and partitions, creating a transparent and open environment. The space is bright and airy, with large windows allowing natural light to fill the room. The glass partitions are framed in metal, and the floor is a polished, reflective surface. In the background, several office chairs and tables are visible, suggesting a collaborative workspace.

# **Zone 1: The Process Traps**

# 1. THE "JUST ONE MORE THING" SYNDROME

Scope Creep: Agile environments are meant to be flexible, but without strict boundaries, changing requirements easily crush the original timeline.

## THE FEATURE SNOWBALL



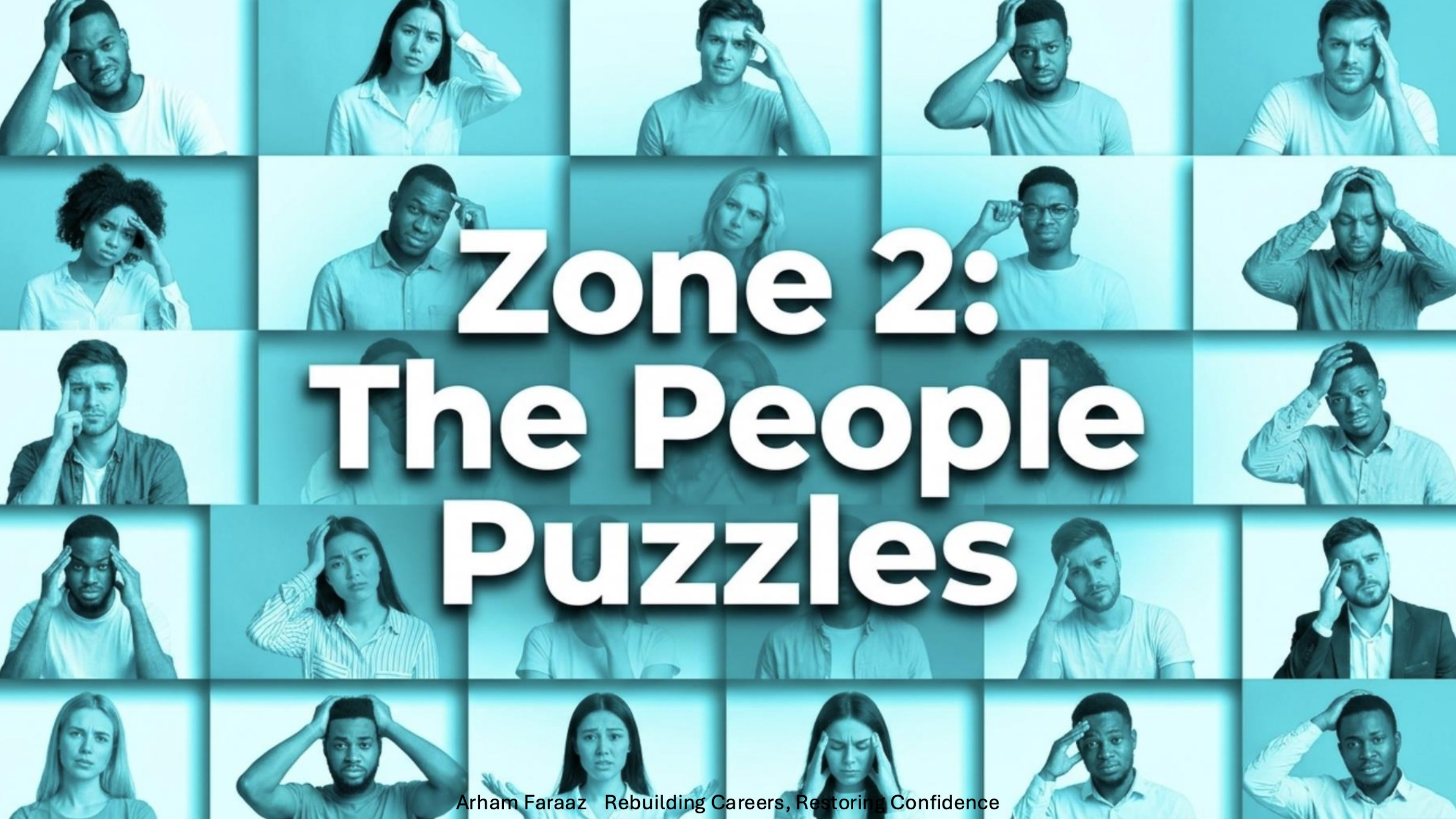
# 2 & 3. Moving Targets and Ticking Clocks

Business &  
Client Pressure



Market &  
Tech Changes

External forces guarantee friction. Whether it's sudden demands for faster delivery (leading directly to burnout?) or unexpected market shifts like new compliance rules, the plan is always under pressure.



# Zone 2: The People Puzzles

# 4. The Remote Reality



**Communication Gaps:** Global and remote teams amplify misunderstandings. When teams aren't perfectly synced, critical details fall through the cracks between time zones.

# 5. Speaking Different Languages

Stakeholder misalignment happens because business and technical teams are literally playing different games with conflicting win conditions.

## The Business Team

**“We need it shipped tomorrow!”**

Focus: ROI

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Delivery Speed

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New Features

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Market Share

## The Technical Team

**“It needs to not crash!”**

Focus: System Stability

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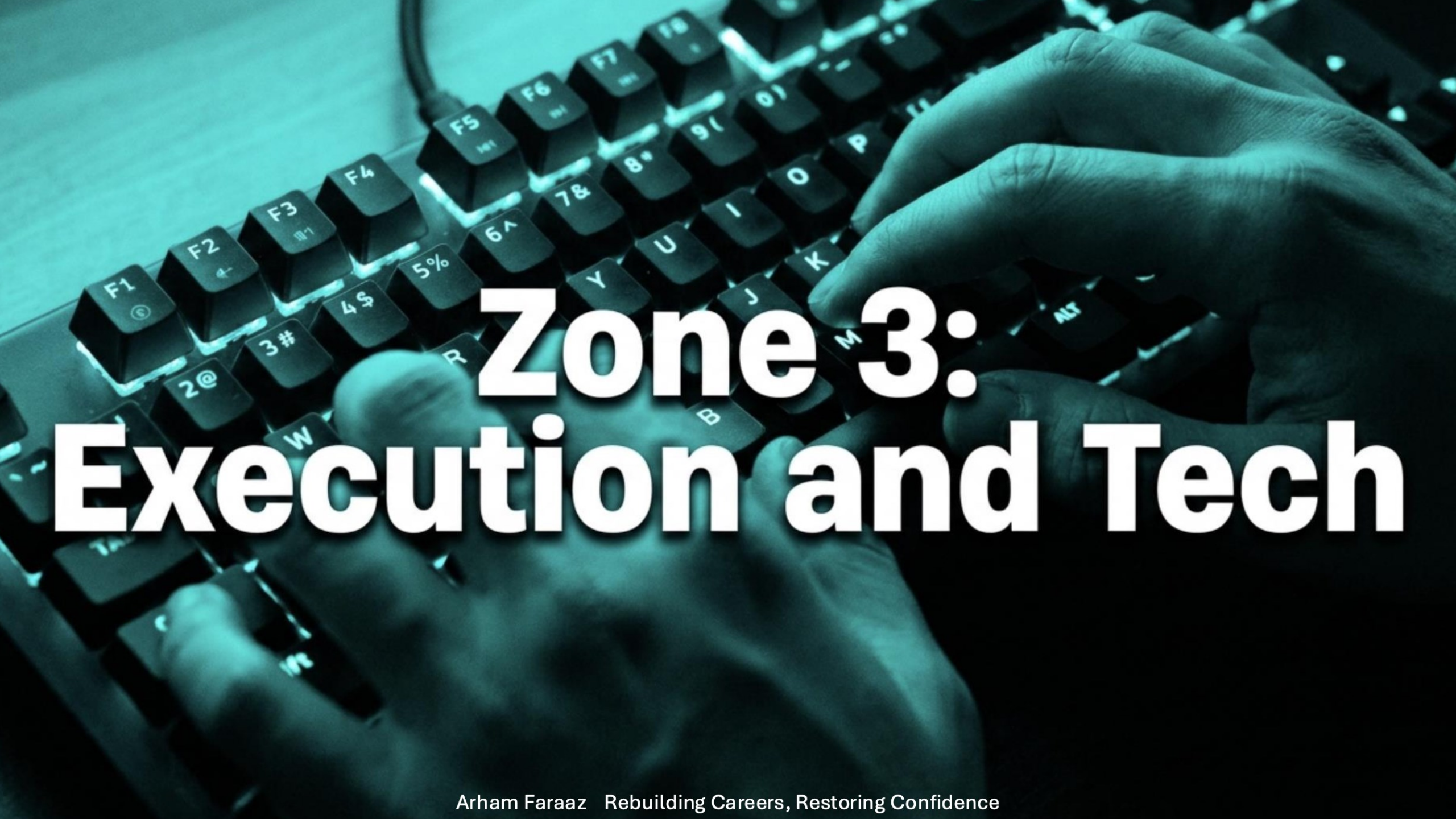
Managing Tech Debt

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Security

---

Clean Code



# Zone 3: Execution and Tech

# 6. The Hidden Monster

Technical Complexity: What looks like a simple button to the user often requires integrating ancient legacy systems and navigating a minefield of technical debt.



**LEGACY BANKING SYSTEMS**

**SPAGHETTI CODE**

**INTEGRATION BUGS**

# 7. Lack of Tools and Talent

Resource Constraints: You cannot deliver at maximum speed without skilled people or the right tools. Pulling on one corner of the triangle inevitably breaks another.



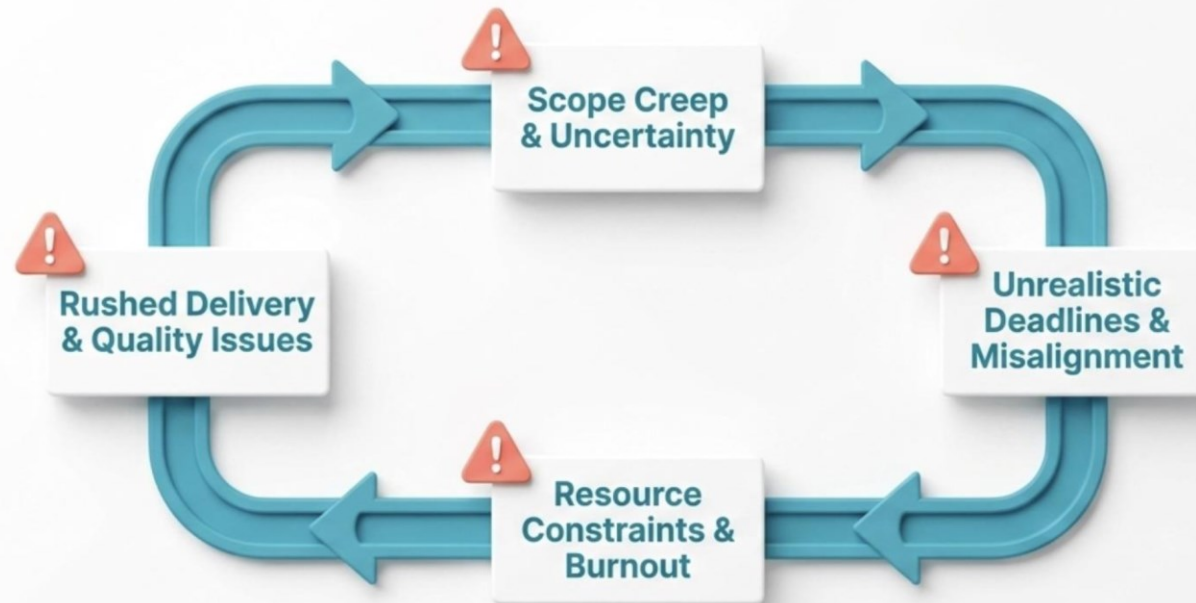
# 8. The Ultimate Fallout

Quality Issues: Poor quality rarely happens on purpose. It is the direct result of everything else going wrong first. Rushed code guarantees bugs, which forces rework, which destroys customer trust.



# The Vicious Cycle of Software Projects

These challenges do not happen in a vacuum. A **failure in the plan inevitably guarantees a failure** in execution. Break the cycle to save the project.



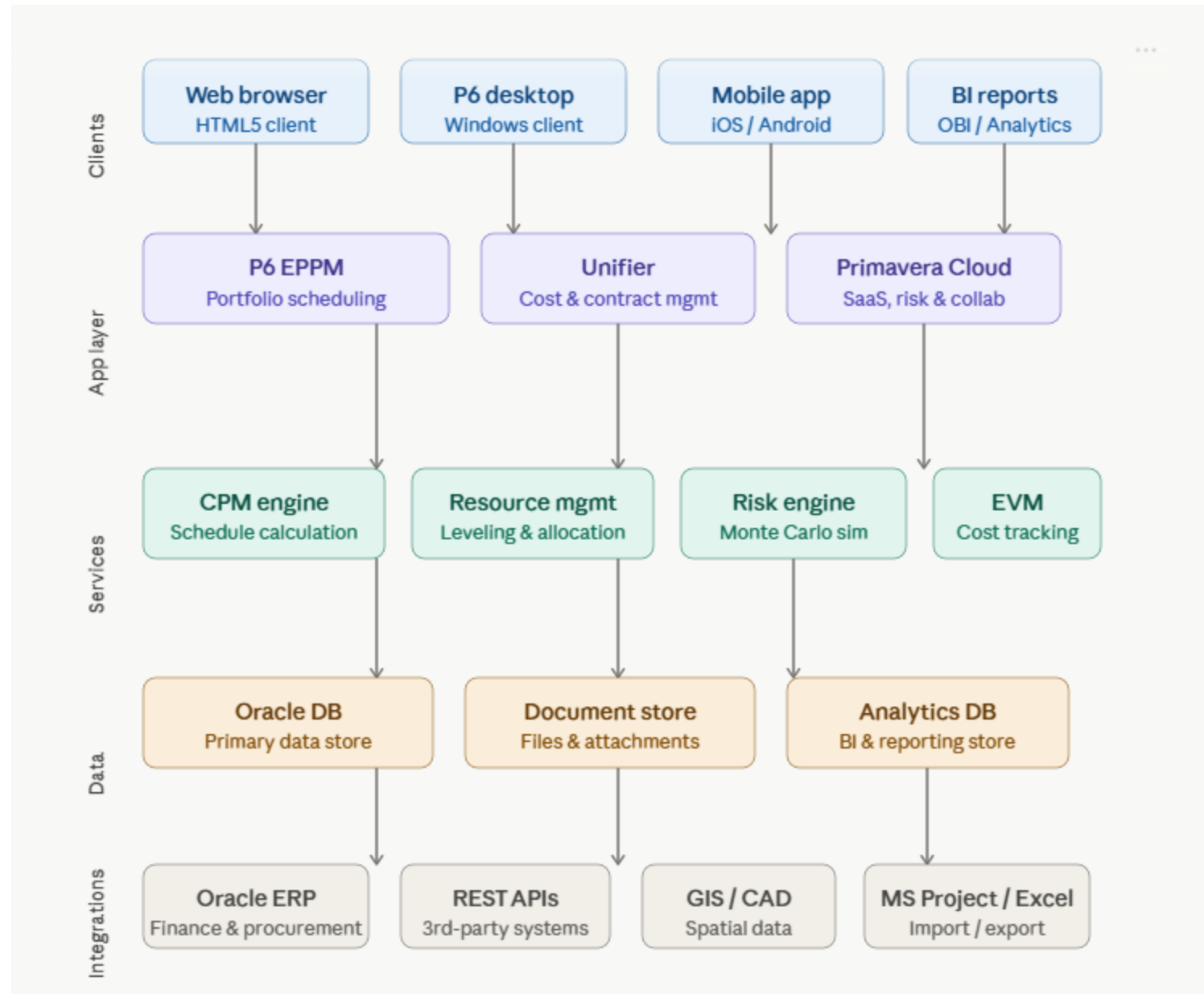
# Software Management Tools - Primavera

- Oracle's Primavera is a comprehensive project portfolio management (PPM) software suite designed for planning, managing, and executing complex, large-scale projects.
- Oracle's Primavera is a comprehensive project portfolio management (PPM) software suite designed for planning, managing, and executing complex, large-scale projects across industries like construction, engineering, oil & gas, aerospace, and utilities.

## Key Components:

- **Primavera P6 EPPM (Enterprise Project Portfolio Management)** — the flagship product, used for scheduling, resource management, and cost control across thousands of activities.
- **Primavera Unifier** — focused on capital program management, contract management, and cost control.
- **Primavera Cloud** — a modern, SaaS-based version offering risk analysis, scheduling, and collaboration tools.

# Software Management Tools - Primavera



# Software Management Tools - Primavera

Layers of Primavera:

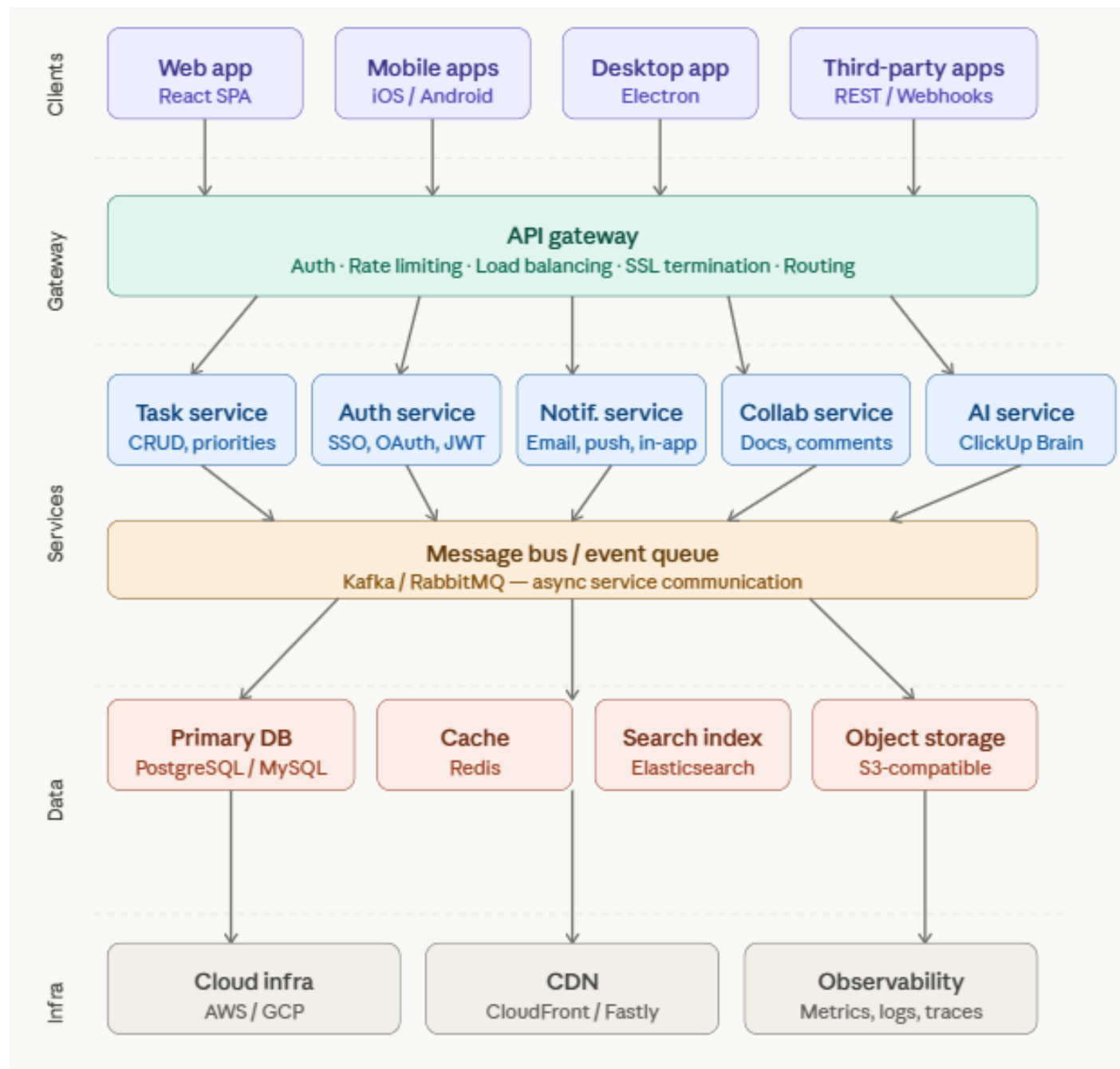
- **Clients** — users access Primavera through a web browser (HTML5), the P6 Windows desktop client, a mobile app, or BI/analytics dashboards.
- **Application layer** — the three main products sit here: P6 EPPM for enterprise scheduling, Unifier for cost and contract management, and Primavera Cloud for the modern SaaS offering.
- **Services** — the core engines that power all products: the CPM (Critical Path Method) scheduler, resource leveling, Monte Carlo risk simulation, and Earned Value Management.
- **Data tier** — an Oracle database for transactional data, a document store for file attachments, and a separate analytics database for reporting.
- **Integrations** — Primavera connects outward to Oracle ERP (finance and procurement), REST APIs for third-party tools, GIS/CAD systems for spatial project data, and MS Project/Excel for import-export workflows.

# Software Management Tools – Click Up

- ClickUp is a cloud-based productivity and project management platform designed to help teams and individuals organize work, collaborate, and manage tasks in one place.

## Key Features:

- **Task Management** – Create, assign, and track tasks with due dates, priorities, and custom statuses.
- **Multiple Views** – Switch between List, Board (Kanban), Gantt, Calendar, and Timeline views to suit different workflows.
- **Docs & Wikis** – Built-in document editor for notes, SOPs, and team knowledge bases.
- **Goals & OKRs** – Set and track targets tied directly to tasks and projects.
- **Automation** – Automate repetitive actions like status changes, assignments, and notifications.
- **Integrations** – Connects with tools like Slack, Google Drive, GitHub, Zoom, and 1,000+ others
- **Time Tracking** – Log time natively or via integrations for billing and productivity insights.
- **AI Features (ClickUp Brain)** – AI assistant for writing, summarizing, and generating project insights.



# Drawbacks of Software Project Management

## 1. **Costs are High**

Consider spending money on various kinds of project management tools, software, & services if ones engage in Software Project Management strategies. These initiatives can be expensive and time-consuming to put in place. Because your team will be using them as well, they may require training. One may need to recruit subject-matter experts or specialists to assist with a project, depending on the circumstances. Stakeholders will frequently press for the inclusion of features that were not originally envisioned. All of these factors can quickly drive up a project's cost.

## 2. **Complexity will be increased**

Software Project management is a multi-stage, complex process. Unfortunately, some specialists might have a propensity to overcomplicate everything, which can lead to confusion among teams and lead to delays in project completion. Their expressions are very strong and specific in their ideas, resulting in a difficult work atmosphere. Projects having a larger scope are typically more arduous to complete, especially if there isn't a dedicated team committed completely to the project. Members of cross-functional teams may lag far behind their daily tasks, adding to the overall complexity of the project being worked on.

# Drawbacks of Software Project Management

## **3. Overhead in Communication**

Recruits enter your organization when we hire software project management personnel. This provides a steady flow of communication that may or may not match a company's culture. As a result, it is advised that you maintain your crew as small as feasible. The communication overhead tends to skyrocket when a team becomes large enough. When a large team is needed for a project, it's critical to identify software project managers who can conduct effective communication with a variety of people.

## **4. Lack of Originality**

Software Project managers can sometimes provide little or no space for creativity. Team leaders either place an excessive amount of emphasis on management processes or impose hard deadlines on their employees, requiring them to develop and operate code within stringent guidelines. This can stifle innovative thought and innovation that could be beneficial to the project. When it comes to Software project management, knowing when to encourage creativity and when to stick to the project plan is crucial. Without Software project management personnel, an organization can perhaps build and ship code more quickly. However, employing a trained specialist to handle these areas, on the other hand, can open up new doors and help the organization achieve its objectives more quickly and more thoroughly.